

Dien Wu

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Education

University of Utah	Ph.D. in Atmospheric Sciences Advisor: Prof. John C. Lin	2016 – 2020	Salt Lake City, US
University of Utah	M.S. in Atmospheric Sciences Advisor: Prof. John C. Lin	2014 – 2016	Salt Lake City, US
Florida State University	B.S. in Meteorology Advisor: Prof. Jon E. Ahlquist	2012 – 2014	Tallahassee, US
Nanjing University of Information Science and Technology	B.S. in Meteorology	2010 – 2012	Nanjing, China

Employment

Assistant Professor	Dept. of Atmospheric Science, Colorado State University	Jan 2025 – present
Staff Scientist (Supervisor: Prof. Paul Wennberg & Dr. Junjie Liu)	Division of Geological and Planetary Sciences, California Institute of Technology	July 2023 – Dec 2024
Postdoctoral Scholar Research Associate (Supervisor: Prof. Paul Wennberg & Dr. Junjie Liu)	Division of Geological and Planetary Sciences, California Institute of Technology	July 2020 – July 2023
Graduate Research Assistant	Dept. of Atmospheric Sciences, University of Utah	Aug 2014 – June 2020

Professional Experience

NASA Tropospheric Emissions: Monitoring of Pollution (TEMPO) Science Team Member	2025 - present
NASA Orbiting Carbon Observatory (OCO-2/3) Science Team Member	2016 – present
Co-convenor of Global Environmental Change session “Attributing, Projecting, and Linking Greenhouse Gas Emissions to Sources, Air Quality and Climate Impacts.” & “Forcings, Responses, and Feedback in the Earth System”	AGU 2023, 2024

List of Peer-Reviewed and In-prep Publications [[Google Scholar](#)]

In review or in preparation (all manuscripts listed below are available upon request)

- 30. Wu, D.,** Liu, J., Suselj, K., Baliaka, H., Crounse, J. D., Li, F., Urbanski, S., Ng, N. L., and Wennberg, P. O.: Observation-constrained Emissions from a Wildland-Urban Interface Fire: the Eaton Fires. Submitted to *Environmental Science & Technology*.
29. Shen, J., Banach, C., Chou, Y., Zawadowicz, M., Taghvaei, S., Sun, H., Campbell, S., **Wu, D.,** Paulson, S.: Dependence of OH formation on biomass burning aerosols and aqueous processing: integrated insights from measurements and modeling, in review under *PNAS*.
- 28. Wu, D.,** Wennberg, P., Liu, K., Liu, J., Anderson, M., and Cawse-Nicholson, K.: Irrigation-based Evapotranspiration over the San Joaquin Valley, In prep for *Agricultural and Forest Meteorology*
- 27. Wu, D.,** Liu, J., Laughner, J. L., Hasheminassab, S., Miyazaki, K., and Wennberg, P. O.: Satellite-based multi-tracer inversion in detecting urban emission variability, In prep for *Atmospheric Chemistry and Physics*.

Published

26. Baliaka, H., Ward, R., Raffuse, S., Ries, B., Cui, Y., **Wu, D.,** Liu, R., Barut, C., Hasheminassab, C., Tissot, F., Russell, A., Seinfeld, J., Flagan, R., Bahreini, R., Wennberg, P. O., and Ng, N.: Fire at the boundary: airborne toxic metals from the 2025 Los Angeles urban firestorm. *Science Advances*, 2026.

25. Xiao, Z., Jones, D. B. A., **Wu, D.**, Kim, J., Nassar, R., & Blanco, B.: High-resolution modeling to quantify CO₂ emissions from an industrial point source. *Journal of Geophysical Research: Atmospheres*, 131, e2025JD045978. <https://doi.org/10.1029/2025JD045978>, 2026.
24. Madsen-Colford, S., Hutyra, L., Smith, I., **Wu, D.**, Arain, M. A., Staebler, R., Ma, W., Restrepo-Coupe, N., and Wunch, D.: Modification and comparison of two urban vegetation models over Southern Ontario, Canada. *Journal of Geophysical Research: Biogeosciences*, 130, e2025JG008943. <https://doi.org/10.1029/2025JG008943>, 2025.
23. Johnson, M. S., Hamilton, S. D., Jeong, S., Cui, Y. Y., **Wu, D.**, Turner, A., and Fischer, M.: State-wide California 2020 carbon dioxide budget estimated with OCO-2 and OCO-3 satellite data, *Atmos. Chem. Phys.*, 25, 8475–8492, <https://doi.org/10.5194/acp-25-8475-2025>, 2025.
22. Jeong, S., Hamilton, S.D., Johnson, M.S., **Wu, D.**, Turner, A.J. and Fischer, M.L.: Applying Gaussian Process Machine Learning and Modern Probabilistic Programming to Satellite Data to Infer CO₂ Emissions. *Environmental Science & Technology*, 59(9), pp.4376-4387, 2025.
21. Hamilton, S.D., **Wu, D.**, Johnson, M.S., Turner, A.J., Fischer, M.L., Dadheech, N. and Jeong, S.: Estimating carbon dioxide emissions in two California cities using Bayesian inversion and satellite measurements. *Geophysical Research Letters*, 51(20), p.e2024GL111150, <https://doi.org/10.1029/2024GL111150>, 2024.
20. Wilmot, Y. T., Lin C. J., **Wu, D.**, Oda, T., and Kort A. E.: Toward a satellite-based monitoring system for urban CO₂ emissions in support of emission reduction targets, *Environ. Res. Lett.*, <https://iopscience.iop.org/article/10.1088/1748-9326/ad6017>, 2024.
19. Madsen-Colford, S., **Wu, D.**, Halim, M. A., Wunch D.: CO₂ Fluxes of Vegetation in the Greenbelt of Ontario and Ecosystem Emissions Associated with its Removal, *Elem. Sci. Anth.*, 12, <https://doi.org/10.1525/elementa.2023.00102>, 2024.
18. Li, M., Kort, E.A., Bloom, A.A., **Wu, D.**, Plant, G., Gerlein-Safdi, C. and Pu, T.: Underestimated Dry Season Methane Emissions from Wetlands in the Pantanal. *Environ. Sci. Technol.* 58, 7, 3278–3287, <https://doi.org/10.1021/acs.est.3c09250>, 2024.
17. **Wu, D.**, Laughner, J. L., Liu, J., Palmer, P. I., Lin, J. C., and Wennberg, P. O.: A simplified non-linear chemistry transport model for analyzing NO₂ column observations: STILT-NO_x, *Geosci. Model Dev.*, 16, 6161–6185, <https://doi.org/10.5194/gmd-16-6161-2023>, 2023.
16. Zazzeri, G., Graven, H., Xu, X., Saboya, E., Blyth, L., Manning, A.J., Chawner, H., **Wu, D.**, and Hammer, S.: Radiocarbon measurements reveal underestimated fossil CH₄ and CO₂ emissions in London. *Geophys. Res. Lett.*, 50(15), p.e2023GL103834, <https://doi.org/10.1029/2023GL103834>, 2023.
15. Mallia, D.V., Mitchell, L.E., Gonzalez Vidal, A.E., **Wu, D.**, Kunik, L. and Lin, J.C.: Can We Detect Urban-Scale CO₂ Emission Changes Within Medium-Sized Cities? *J. Geophys. Res.-Atmos.*, 128(11), p.e2023JD038686, <https://doi.org/10.1029/2023JD038686>, 2023.
14. Wu, K., Palmer, P. I., **Wu, D.**, Jouget, D., Feng, L., and Oda, T.: Theoretical assessment of the ability of the MicroCarb satellite city-scan observing mode to estimate urban CO₂ emissions, *Atmos. Meas. Tech.*, 16, 581–602, <https://doi.org/10.5194/amt-16-581-2023>, 2023.
13. **Wu, D.**, Liu, J., Wennberg, P. O., Palmer, P. I., Nelson, R. R., Kiel, M., and Eldering, A.: Towards sector-based attribution using intra-city variations in satellite-based emission ratios between CO₂ and CO, *Atmos. Chem. Phys.*, 22, 14547–14570, <https://doi.org/10.5194/acp-22-14547-2022>, 2022.
12. Roten, D., Lin, J. C., Kunik, L., Mallia, D., **Wu, D.**, Oda, T., and Kort, E. A.: The Information Content of Dense Carbon Dioxide Measurements from Space: A High-Resolution Inversion Approach with Synthetic Data from the OCO-3 Instrument, *Atmos. Chem. Phys. Discuss.* [preprint], <https://doi.org/10.5194/acp-2022-315>, in review, 2022.
11. Lei, R., Feng, S., Danjou, A., Broquet, G., **Wu, D.**, Lin, J.C., O'Dell, C.W. and Lauvaux, T.: Fossil fuel CO₂ emissions over metropolitan areas from space: A multi-model analysis of OCO-2 data over Lahore, Pakistan. *Remote Sens. Environ.*, 264, p.112625, <https://doi.org/10.1016/j.rse.2021.112625>, 2021.

10. **Wu, D.**, Lin, J. C., Duarte, H. F., Yadav, V., Parazoo, N. C., Oda, T., and Kort, E. A.: A model for urban biogenic CO₂ fluxes: Solar-Induced Fluorescence for Modeling Urban biogenic Fluxes (SMURF v1), *Geosci. Model Dev.*, 14, 3633–3661, <https://doi.org/10.5194/gmd-14-3633-2021>, 2021.
9. Qu, Z., **Wu, D.**, Henze, D. K., Li, Y., Sonenberg, M., and Mao, F.: Transboundary transport of ozone pollution to a US border region: a case study of Yuma. *Environmental Pollution*, 273, Pp. 116421, <https://doi.org/10.1016/j.envpol.2020.116421>, 2021.
8. Roten, D., **Wu, D.**, Fasoli, B., Oda, T., & Lin, J. C.: An interpolation method to reduce the computational time in the Stochastic Lagrangian particle dispersion modeling of spatially dense XCO₂ retrievals. *Earth and Space Science*, 8, e2020EA001343. <https://doi.org/10.1029/2020EA001343>, 2021.
7. Ye, X., T. Lauvaux, E.A. Kort, T. Oda, S. Feng, J.C. Lin, E. Yang, and **Wu, D.**: Constraining fossil fuel CO₂ emissions from urban area using OCO-2 observations of total column CO₂. *J. Geophys. Res.-Atmos.*, 125, e2019JD030528. <https://doi.org/10.1029/2019JD030528>, 2020.
6. Yang, E.G., E.A. Kort, Wu, D., Lin, J., Oda, T., Ye, X., and Lauvaux, T.: Using space-based observations and Lagrangian modeling to evaluate urban carbon dioxide emissions in the Middle East. *J. Geophys. Res.-Atmos.*, 125, e2019JD031922. <https://doi.org/10.1029/2019JD031922>, 2020.
5. **Wu, D.**, J.C. Lin, T. Oda, and E.A. Kort: Space-based quantification of per capita CO₂ emissions from cities, *Environ. Res. Lett.*, <https://doi.org/10.1088/1748-9326/ab68eb>, 2020. ([NASA media coverage](#))
4. Hernandez, A.J., Morales-Rincon, L.A., **Wu, D.**, Mallia, D., Lin, J.C. and Jimenez, R.: Transboundary transport of biomass burning aerosols and photochemical pollution in the Orinoco River Basin. *Atmos. Environ.*, <https://doi.org/10.1016/j.atmosenv.2019.01.051>, 2019.
3. **Wu, D.**, Lin, J. C., Fasoli, B., Oda, T., Ye, X., Lauvaux, T., Yang, E. G., and Kort, E. A.: A Lagrangian approach towards extracting signals of urban CO₂ emissions from satellite observations of atmospheric column CO₂ (XCO₂): X-Stochastic Time-Inverted Lagrangian Transport model (“X-STILT v1”), *Geosci. Model Dev.*, 11, 4843–4871, <https://doi.org/10.5194/gmd-11-4843-2018>, 2018.
2. Mallia, D.V., Kochanski, A., Wu, D., Pennell, C., Oswald, W., and Lin, J.: Wind-Blown Dust Modeling Using a Backward-Lagrangian Particle Dispersion Model. *J. Appl. Meteor. Climatol.*, 56, 2845–2867, <https://doi.org/10.1175/JAMC-D-16-0351.1>, 2017.
1. Lin, J. C., Mallia, D. V., **Wu, D.**, and Stephens, B. B.: How can mountaintop CO₂ observations be used to constrain regional carbon fluxes? *Atmos. Chem. Phys.*, 17, 5561–5581, <https://doi.org/10.5194/acp-17-5561-2017>, 2017.

Other Documentations

2. Contributing authors of three sections on GHG implications to the NASA white paper, 2026.
1. Turnbull, J. C., Curras, T., Gurney, K. R., Hilton, T. W., Mueller, K. L., Vogel, F., Yao, B., Albarus, I., Ars, S., Baidar, S., Chatterjee, A., Chen, H., Chen, J., Christen, A., Davis, K. J., Hajny, K., Han, P., Karion, A., Kim, J., Lopez Coto, I., Papale, D., Ramonet, M., Sperlich, P., Vardag, S. N., Vermeulen, A., Vimont, I. J., **Wu, D.**, Zhang, W., Augusti-Panareda, A., Ahlgren, K., Ahn, D., Boyle, T., Brewer, A., Brunner, D., Cai, Q., Chambers, S., Chen, Z., Dadheech, N., D’Onofrio, C., Dunse, B. L., Engelen, R., Fathi, S., Gioli, B., Hammer, S., Hase, F., Hong, J., Hutyrá, L. R., Järvi, L., Jeong, S., Karstens, U., Kenion, H. C., Kljun, N., Laurent, O., Lauvaux, T., Lin, J. C., Liu, Z., Loh, Z., Maier, F., Matthews, B., Mauder, M., Miles, N., Mitchell, L., Monteiro, V. C., Mostafavi Pak, N., Röckmann, T., Roiger, A., Roten, D., Scheutz, C., Shahrokhi, N., Shepson, P. B., Stagakis, S., Tong, X., Trudinger, C. M., Velasco, E., Whetstone, J. R., Winbourne, J. B., Wu, J., Xueref-Remy, I., Yadav, V., Yu, L., Zazzeri, G., Zeng, N., and Zhou, M.: IG3IS Urban Greenhouse Gas Emission Observation and Monitoring Good Research Practice Guidelines WMO GAW Report, World Meteorological Organisation, Geneva Switzerland, 2025.

Current Research Grants and Roles

Aug 2025 –	NASA A.21 TEMPO/ACX Science and Applications Teams: Quantifying NO _x	PI
July 2028	emissions from agricultural lands (Co-Is: Stephen Ogle, Anping Chen, CSU)	(\$566k/3yrs)

Oct 2026

Oct 2024 – Sept 2027	NASA A.6 Carbon Monitoring System: Towards a Multi-scale Near Real-Time Carbon Monitoring System for Vegetation Fires (CMS-Fire) (PI: Dr. Junjie Liu, JPL)	Institutional PI (\$1M/3yrs across JPL and CSU ATS)
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Past Research Grants

Aug 2021 – July 2024	NASA A.51 Science Team for the OCO Missions: Beyond quantification of carbon emissions from global cities: attribution to socioeconomic sectors using the OCO missions and multiple remote sensing products (PI: Prof. Eric Kort, UMichigan)	Co-I
2021 - 2023	NASA Research and Technology Development Fund: Quantification of Urban Emissions using OCO-3 Snapshot Area Maps (PI: Dr. Annmarie Eldering, NIST)	Co-I
June 2021 – May 2024	NASA A.5 Carbon Cycle Science: 2020 California Carbon Dioxide Budget in a Changing Environment (PI: Dr. Seongeun Jeong, LBNL)	Collaborator

Invited Workshops

2.	OCO Science and Application Center brainstorming workshop @Caltech	Sept 2-3, 2026
1.	Keck Workshop: Forging Community Consensus for an Integrated GHG and Winds Program @Caltech	Oct 7 – 11, 2024

Invited Seminars/Talks

13.	AMS27 Invited Talk at Session: <i>Air Quality Insights from Satellite Observations</i> (Conveners: Pieternel Levelt, Gonzalo Gonzalez Abad, Hyeong-Ahn Kwon, Wenfu Tang, and Ivan Ortega)	Jan 2027
12.	AGU26 Invited Talks at two Sessions: <i>A064: Bridging the Gap: Integrating Multi-Platform Observations of Trace Gases in Earth's Atmosphere</i> <i>A048: Biosphere–Atmosphere Chemical Interactions Across Land and Ocean Systems in a Changing Earth</i>	Dec 2026
11.	Seminar at the School of Marine and Atmospheric Sciences, <i>Stony Brook University</i> https://www.stonybrook.edu/somas/news/seminars.html (host: Prof. Guanyu Huang)	Nov 18, 2026
10.	Atmospheric Chemistry Observations & Modeling (ACOM) seminar, <i>NCAR</i> https://www2.acom.ucar.edu/seminars (host: Dr. Wenfu Tang)	April 30, 2026
9.	Seminar at air quality big group in the Dept of Mechanical Engineering, <i>University of Colorado at Boulder</i> (host: Prof. Daven Henze)	Nov 13, 2025
8.	Remote Seminar at the Dept of Ecological and Environmental Informatics Research, <i>Northern Arizona University</i> (host: Prof. Andrew Richardson)	Oct 20, 2025
7.	Seminar at the Dept of Civil and Environmental Engineering, <i>Colorado State University</i> , https://www.engr.colostate.edu/ce/events/water-seminar/ (host: Prof. Frances Davenport)	Oct 8, 2025
6.	Seminar at the Dept of Atmospheric Science, <i>University of Wyoming</i> , https://www.uwyo.edu/atasc/seminars.html (host: Prof. Dana Caulton)	April 8, 2025
5.	CLASP seminar at the <i>University of Michigan</i> (host: Prof. Adriana Bailey)	Nov 21, 2024
4.	BASC Fall 2024 seminar at the <i>University of California, Berkeley</i> , https://atmosphere.berkeley.edu/seminars (host: Prof. Paolo D'Odorico)	Oct 2, 2024
3.	Colloquia at the Dept. of Atmospheric Science, <i>Colorado State University</i> , https://www.atmos.colostate.edu/colloquia/?term=sp2024 (host: Prof. Jeffrey Pierce)	Feb 15, 2024
2.	Annual seminar series at Institute of Atmospheric Physics, <i>Chinese Academy of Sciences</i>	June 2023
1.	Noble seminar series, Earth, Atmospheric, and Planetary Physics, <i>the University of Toronto</i> , https://www.physics.utoronto.ca/research/eapp/atmospheric-physics-seminars/dien/	Oct 31, 2022

Current Group Members

	Oct 2026
Olivia Griffith (MS student)	Fall 2026 -
Molly McKellar (MS student)	Fall 2026 -
Megan Marra (MS student)	Fall 2025 - present
Jessie Lyons (co-advised MS student, funded/advised by Dr. Andrew Schuh)	Spring 2025 - present

Teaching and mentoring experiences

Classes @CSU: <i>Land-Atmosphere Interaction</i> (ATS 761)	Fall 2025
Guest lecture, Course on GIS & RS, @NYU (~75 mins, 20+ undergrads) “Remote sensing of trace gases and aerosols”	April 2024
Guest lecture, ESE 104, @Caltech “Seeing Urban Emissions through Satellite Lenses”	Feb 2024
Caltech’s Center for Teaching, Learning, and Outreach (CTLO) workshop “Transforming your research into classes”	Spring, Summer 2023
Kevin Do (undergraduate student, Dept. of Computer Sciences, Caltech) Project: <i>Estimation of global CO₂ mass using TCCON</i>	2023
Zixuan Xiao (graduate student, Dept. of Physics, University of Toronto) Project: <i>Power plant CO₂ emission estimates using X-STILT</i>	2023
Dr. Ariana Tribby (former graduate student, Dept. of Env Sci & Eng, Caltech) Project: <i>CH₄ emission estimates for oil & gas using TCCON</i>	2023

MS and PhD Dissertation Committees

Internal Ph.D. Committees @ Dept. of Atmospheric Science at CSU

James Larson (advised by Prof. Jim H.)
 Brandon McGuire (advised by Profs. Emily F. and Jeff P.)
 Camille Mavis (advised by Prof. Sonia K.)
 Leif Fredericks (advised by Prof. Maria R.)
 Daniela Guevara (advised by Prof. Emily F.)
 Lena Low (advised by Prof. Emily F.)

External Ph.D. Committee

Ajay Devda (advised by Prof. Jessica O’Connell), Dept. of Ecosystem Science and Sustainability at CSU

Peer Review Activities

Proposal Reviewer

NOAA [Atmospheric Chemistry, Carbon Cycle and Climate \(AC4\)](#) Call (FY23, ad hoc review)
 NSF [Atmosphere Cluster \(AC\)](#) Call (FY26, ad hoc review)

Paper Reviewer

Science Advances, Geophysical Research Letters (GRL), Environmental Research Letters (ERL), Remote Sensing of Environment (RSE), Environmental Science & Technology and Letters (EST, ESTL), Atmospheric Chemistry and Physics (ACP), Atmospheric Measurement Techniques (AMT), Biogeosciences (BG), Journal of Geophysical Research: Atmospheres, Biogeosciences (JGR-A, -B), Earth’s Future, Scientific Reports, Elementa: Science of the Anthropocene, and Ecosphere.

Selected Presentations (Group members are *underlined*)

Wu, D. , Laughner, J., Ogle, S., Wang, Z., Chen, A., and Griffith, O.: Towards Constraining Soil NO _x Fluxes and the Responses to Soil Moisture from TEMPO, 29th Conference on Atmospheric Chemistry, 107th AMS Annual Meeting, 2027 (oral)	Jan 10 – 17, 2027
<u>*Marra, M.*</u> , Liu, J., and Wu, D. : Assessing Confidence in Evaluating Fire Emission Inventories: Impacts from Near-Field Turbulence, Fire-Induced Weather, and Plume Rise (oral)	Dec 7 – 11, 2026

Wu, D. , Liu, J., Suselj, K., Baliaka, H., Crounse, J. D., Li, F., Urbanski, S., Marra, M., Ng, N. L., and Wennberg, P. O.: Observation-constrained PM _{2.5} and Pb Emissions from the Eaton Fire at the Wildland–Urban Interface, <i>AGU Fall Meeting</i> , 2026 (oral)	Dec 7 – 11, 2026
Wu, D. , Laughner, J., Ogle, S., Wang, Z., Chen, A., and Griffith, O.: Towards Constraining Soil NO _x Fluxes and the Responses to Soil Moisture from TEMPO, <i>AGU Fall Meeting</i> , 2026 (invited talk)	Dec 7 – 11, 2026
Wu, D. , Towards quantifying NO _x emissions from agricultural lands, TEMPO science team meeting, Iowa City, IA (oral)	June 15 – 18, 2025
Wu, D. , Suselj, K., Crounse, J., Baliaka, H., Urbanski, S., Mallia, D., Holder, A., Wennberg, P., and Liu, J., Investigating Emissions of CO and PM _{2.5} from an Urban Fire: the Eaton Fires, <i>AGU Fall Meeting</i> , New Orleans (author requested poster)	Dec 14-19, 2025
Wu, D. , Towards quantifying NO _x emissions from agricultural lands, TEMPO science team meeting, Cambridge, Massachusetts (oral)	Aug 19-22, 2025
Wu, D. , Wennberg, P., Liu, K., Liu, J., and Cawse-Nicholson, K.: Quantifying Irrigation Signals over Orchard Ranches in San Joaquin Valley using Ranch-scale and Space-based Observations, <i>AGU Fall Meeting</i> , Washington DC (oral)	Dec 9-13, 2024
Wu, D. , Liu, J., Laughner, J. L., and Wennberg, P. O.: Towards monitoring, reporting, and verifying urban emissions from space, <i>20th International Workshop on Greenhouse Gas Measurements from Space (IWGGMS-20)</i> , Boulder, CO (oral)	May 29 – 31, 2024
Wu, D. , Cawse-Nicholson, K., Liu, J., Duarte, F. H., Wennberg, P.: Interpreting Atmospheric Imprints of Intensive Agricultural Activities using Multiple Satellites, <i>AGU Fall Meeting</i> , San Francisco, CA (oral)	Dec 11-15, 2023
Wu, D. , Liu, J., Wennberg, P., Palmer, P. I., Nelson, R. R., Laughner, J. L., and Eldering, A.: Informing intra-city emission characteristics using satellite observations of CO ₂ and co-emitted species, <i>AGU Fall Meeting</i> , Chicago, IL (poster)	Dec 12-16, 2022
Wu, D. , J.C. Lin, B. Fasoli, T. Oda, E.A. Kort, and Duarte, H., Towards quantifying urban CO ₂ emissions for global cities using column CO ₂ and SIF data, <i>AGU Fall Meeting</i> , San Francisco, CA (eLightning)	Dec 09-13, 2019
Wu, D. , J.C. Lin, H. Duarte, G. Wei, K. Wu, S. Richardson, N. Miles, K. Davis, E. A. Kort: Towards improving the modeling of urban biosphere using Solar-induced Fluorescence (SIF), <i>AGU Chapman Conference on understanding carbon climate feedbacks</i> , San Diego, CA (poster)	Aug 26-29, 2019
Wu, D. , J. C. Lin, Oda, T., Ye, X., Lauvaux, T., Yang, E., and Kort, E. A., Towards Interpreting the Signal of CO ₂ Emissions from Megacities by Applying a Lagrangian Receptor-oriented Model to OCO-2 XCO ₂ data, <i>AGU Fall Meeting</i> , New Orleans, LA (oral)	Dec 11-15, 2017
Wu, D. , D. V. Mallia, S. P. Urbanski, J. C. Lin, Top-down Constraints on CO Emissions from Wildfire Inventories Using a Receptor-oriented Lagrangian Particle Dispersion Model, <i>AMS Third Conference on Biogeoscience</i> , Salt Lake City, UT (oral)	June 20-25, 2016
Lin, J. C., B. B. Stephens, D. V. Mallia, D. Wu , H. Duarte, S. Urbanski, and J. Ehleringer, How can we constrain regional carbon fluxes in the American Rockies from atmospheric measurements? <i>5th NCAP and AmeriFlux Joint Meeting</i> , Washington DC	Jan 26-29, 2015
Lin, J. C., D. V. Mallia, D. Wu , S. Urbanski, and B. B. Stephens, Quantifying the influence of biomass burning on measurements site in the western U.S., <i>AGU Fall Meeting</i> , San Francisco, CA	Dec 15-19, 2014

Technical Skills

Programming languages and skills: R, Fortran, Python, Bash, LaTeX, Machine Learning
 Modeling experience: HYSPLIT-STILT, WRF-ARW, WRF-Chem, WRF-Fire, CESM (prelim)

Model Assets

1. Column-Stochastic Time-Inverted Lagrangian Transport (X-STILT) model
 Github: <https://github.com/uataq/X-STILT> with DOI: [10.5281/zenodo.1241515](https://doi.org/10.5281/zenodo.1241515)
2. Solar-Induced Fluorescence for Modeling Urban biogenic Fluxes (SMUrF) model
 Github: <https://github.com/wde0924/SMUrF> with DOI: [10.5281/zenodo.4018124](https://doi.org/10.5281/zenodo.4018124)
 Data citation: Wu, D., and J.C. Lin. 2021. Urban Biogenic CO₂ fluxes: GPP, R_{eco} and NEE Estimates from SMUrF, 2010-2019. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAAAC/1899>
3. Stochastic Time-Inverted Lagrangian Transport model for NO_x chemistry (STILT-NO_x)
 DOI: <https://zenodo.org/records/8057850>

Professional Association

American Geophysical Union
 American Meteorological Society

Scholarships and Honors

- Edward J. Zipser Award, Excellence in Graduate Research, University of Utah, May 2020
- Pass the Graduate Qualifying Exam with distinction, University of Utah, May 2015
- Graduate with Magna cum laude, Florida State University, May 2014
- Dean's List, Florida State University, Fall 2012, Spring 2013, Fall 2013, Spring 2014
- Prize for being one of the excellent class leaders, NUIST, June 2012
- Second-tier scholarship (top 10%), NUIST, 2010, 2011